Do More Expensive AI Models Offer Better Value?

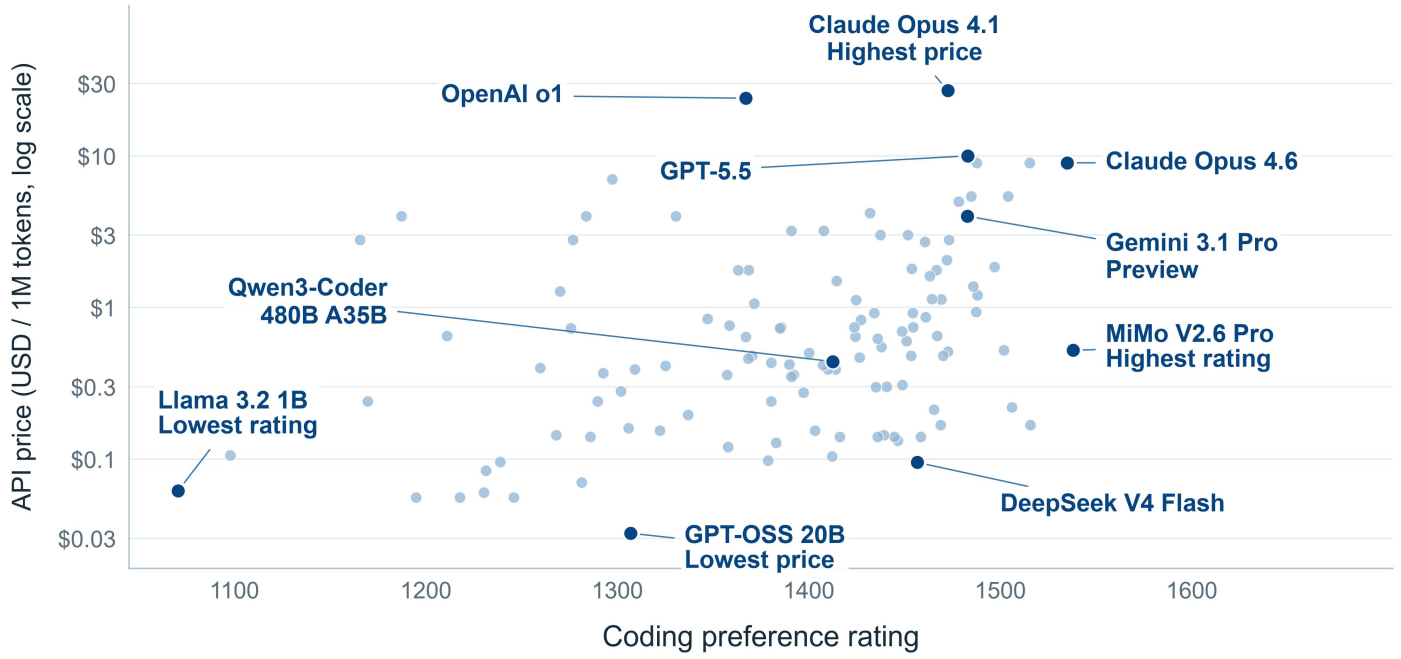
Sicheng Yi

September 29, 2026

01 Question Building

Small game developer team

- Write code
- Review code



Many models. Different API prices. Limited budget.

Team Demand: Do higher API prices mean higher coding ratings and better value for our team?
Statistical Question: Across AI models, are higher API prices associated with higher coding preference ratings?

02 Data Sources

Arena coding

Ratings:
Which coding answers do people prefer?

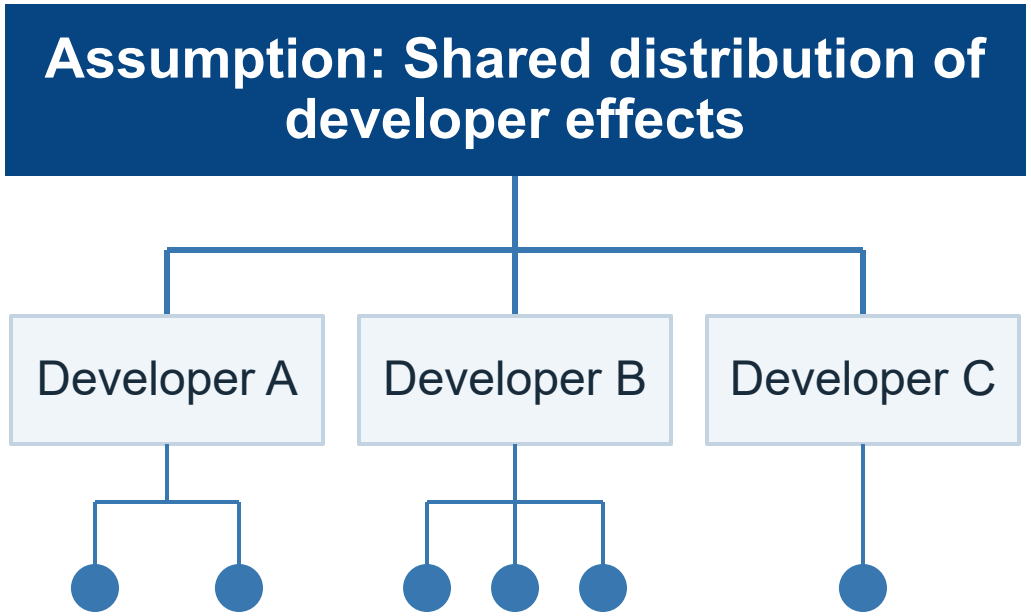
OpenRouter

API Prices:
What do input and output tokens cost?



119 Models with coding ratings and API prices → **23** Model Developers

03 Bayesian Hierarchical Model



Individual models (illustrative structure)

Allow differences

Allow developer-specific baseline ratings.

Share information

Stabilize estimates for developers with fewer models.

Quantify uncertainty

Report credible intervals for price–rating associations.

Compare all developers. Separate within- and between-developer associations.
Related model versions can cause leakage, so models from the same family will stay in the same validation fold.

04 Scope of the Project

Core	**Coding ratings, API prices, one snapshot** Verified matches, hierarchical regression, validation and stability checks
If time allows	Model-family effects or another snapshot
Outside this project	Paid task tests, game-engine benchmarks, self-hosting costs and app development

05 What Counts as A Success Project



Answer the question

Estimate price–rating associations and their uncertainty.



Check the evidence

Assess model fit and sensitivity to key assumptions.



Guide the next test

Shortlist models under stated budget and usage assumptions

Success means a supported answer and a clear next step for the team.

06 Four steps, four checkpoints

Step 1 — Step 2 — Step 3 — Step 4

Match

Match model versions and link coding ratings to API prices.

Analyze

Fit the Bayesian hierarchical model and estimate price–rating associations.

Check

Check **model diagnostics** and assess **sensitivity** to key assumptions.

Deliver

Produce a **model shortlist**, a reproducible **Rmd** report, and analysis code stored in a **github repository**.

07 How I Will Manage the Project

Weekly review and two short updates in Slack

Review the evidence

Check model matches;
Diagnostic plots;
Validation results.

Check reproducibility and stability

Review reproducible code;
Sensitivity to key assumptions.

Document scope changes in Slack

Record the reason for the change;
Proposed action;
Impact on scope and timeline.

Document rating biases and task coverage.
Use public model-level data only.